

Project Report: Mobile Optical LED Communication Receiver

Project Context and Objective

The project objective is to develop a working prototype of a proprietary mobile receiver for a five-bit optical LED signal. The receiver uses a phone camera to detect a physical marker pattern, estimate the state of five LEDs in real time, synchronize to the transmitted optical stream, and reconstruct the original message in a mobile application.

The work has two separate technical layers:

1. a computer vision layer that detects the marker system and converts camera frames into timestamped soft LED observations;
2. a communication and decoding layer that synchronizes the stream and reconstructs the message from noisy, missing, and uncertain observations.

The computer vision layer is necessary for the system to work, but the main technical novelty of the project is in the second layer: a receiver-friendly soft, rateless optical communication protocol with online belief propagation decoding.

Accomplishments and Progress Toward the Goal

Mobile Receiver Prototype

A working Android prototype was implemented using Jetpack Compose and CameraX. The application opens the camera stream, processes frames in real time, displays the detected marker and LED positions, tracks the decoding state, supports scan cancellation, and shows the final decoded binary image message.

The receiver is structured so that the camera and vision components are not tightly coupled to the protocol decoder. The protocol layer only consumes timestamped soft observations for five LED channels, plus marker availability. This separation is important because the vision system can continue to be improved without changing the communication protocol.

Virtual Optical Transmitter

A browser-based virtual transmitter was built to emulate the future physical device. It renders the marker pattern and five LED channels on a screen, supports several transmission rates, and allows a binary image message to be edited and sent. This made it possible to test the full camera-to-decoder pipeline before physical hardware is available.

The virtual transmitter was also useful for protocol iteration. It exposed practical issues that are easy to miss in an abstract model, such as frame timing mismatch, transition frames between LED states, and temporary loss of marker tracking.

Computer Vision Layer

The marker layout was defined as one constrained geometric pattern: a start marker, five fixed LED positions, and an end marker. The receiver estimates a small set of global pose parameters for the whole pattern, such as position, angle, and scale, rather than independently detecting every marker and LED. This keeps the geometry stable and makes the receiver easier to reason about.

Several vision approaches were explored. The first versions used hand-written image scoring and deterministic pose refinement. These versions were useful for validating the geometry and the data flow, but they were sensitive to lighting, reflections, display material, and background clutter. Later experiments moved toward learned likelihood models for marker scoring and LED scoring. These models are still being optimized for mobile speed, so they are not treated here as a final implementation detail. The stable architectural result is that the CV layer provides soft evidence rather than hard binary decisions.

Soft Rateless Fountain Protocol

The most important progress was the shift from simple hard-packet decoding to a soft rateless protocol. Early versions treated each observed LED state as a hard five-bit packet. This was not reliable: a small number of wrong visual decisions could lead to a confidently wrong decoded message. Directly transmitting message bits was also fragile because each source bit depended too much on a specific time interval in the camera stream.

The current protocol uses a Raptor-like structure:

- the source message is protected by a finite precode, conceptually similar to an LDPC precode;
- the transmitted stream is an ongoing sequence of sparse LDGM-style XOR measurements of the protected bits;
- each optical packet carries five measurement bits through the five LEDs;
- the receiver keeps soft likelihoods instead of immediately converting observations into hard bits;
- the message is reconstructed online with belief propagation.

This is a receiver-friendly design. A fixed block-code transmitter would blink for a fixed amount of time and then stop. If the user moved the phone or lost several frames, the receiver could be forced to restart from zero. With the rateless design, the transmitter can keep producing useful measurements, and the receiver can keep the progress already accumulated. A short observation gap slows convergence instead of invalidating the whole scan.

Preamble, Clock Synchronization, and Sampling

The receiver uses a known preamble before the payload. The preamble has two roles. First, it indicates that a new stream has started. Second, it allows the receiver to estimate the stream timing: start time, symbol period, and phase. This is required because the transmitter update rate and the phone camera frame rate are independent clocks.

After preamble acquisition, packets are sampled according to the inferred symbol clock. The receiver does not simply use the nearest raw camera frame. It can aggregate LED observations inside a timing window around the expected symbol center and avoid transition-heavy regions where the displayed LEDs may be between states. The output of this stage is a soft packet observation or an erasure, not just a hard five-bit word.

Online Belief Propagation Decoder

The online decoder maintains a factor graph. Variable nodes represent protected message bits. Each received five-bit packet adds sparse XOR measurement factors to the graph. Belief propagation updates soft messages on graph edges and produces posterior beliefs for the message bits.

This differs from a conventional “receive a block, then decode” workflow. The decoder runs during reception. New observations are incorporated as they arrive, and the current posterior can be shown to the user as a gradually appearing binary image. Uncertain pixels remain gray; confident pixels move toward black or white. Decoding completes when the posterior confidence and code consistency are sufficient, not when a fixed number of packets has been received.

This online soft-decoding approach is the core reason the system can tolerate imperfect vision. Better tracking and LED scoring still improve performance, but the protocol no longer assumes that every frame produces a correct hard packet.

Significant Problems and Solutions

The Project Changed From “Read LEDs” to “Build a Communication Receiver”

The original task looked mostly like a computer vision problem: find the marker, read five LEDs, and decode their state. During prototyping it became clear that this model was too simple. A phone camera observes an asynchronous optical stream, not an ideal digital signal. Some frames are captured near LED transitions, some observations are weak, and some symbols are missing.

The solution was to redesign the receiver as a soft communication system. The CV layer produces likelihoods, the timing layer samples those likelihoods according to the inferred symbol clock, and the decoder reconstructs the message from soft linear measurements.

Hard Decisions Were Too Fragile

Hard per-frame decoding was tested and proved unreliable. Once LED scores were converted too early into hard bits, the decoder lost information about uncertainty. Worse, incorrect hard packets could make the final message wrong with high confidence.

The solution was to preserve soft information through the receiver pipeline. LED observations are kept as scores/likelihoods, uncertain packets can become erasures, and the belief propagation decoder uses this uncertainty instead of pretending that every packet is equally reliable.

Direct Transmission Was Too Sensitive to Local Errors

A simple direct transmission scheme makes specific message bits depend on specific moments in time. If those moments are corrupted by a bad frame, a transition, or a temporary tracking loss, the corresponding message bits are damaged directly.

The solution was to transmit sparse linear combinations of protected bits. This spreads information across many measurements. A weak observation usually reduces confidence rather than destroying a particular source bit. Additional packets can continue improving the posterior belief.

Fixed-Length Transmission Was Bad for User Experience

In a mobile camera workflow, the user may briefly move the phone, lose the marker, or miss several frames. With a fixed block transmission, that can require a full restart. This is inconvenient and makes the system feel unstable.

The solution was to switch to a rateless model. The receiver can preserve already accumulated progress and continue decoding when usable observations resume. This is especially important for a handheld receiver, where short interruptions are normal.

Computer Vision Robustness and Performance

The vision layer had to handle reflections, different screen materials, lighting changes, and background clutter. Hand-written scoring was not robust enough in all cases. Learned likelihood models are being explored, but mobile inference speed remains an important constraint.

The architectural solution was to decouple CV from the protocol. The decoder does not depend on a specific tracker implementation. A better tracker or LED scorer can be substituted later, and the same soft rateless protocol will continue to work. Improving CV should mainly make the decoder converge faster and fail less often.

Outcomes for the Company

The project produced a working prototype and a clear technical direction for a proprietary mobile optical receiver. The company can use these results as the basis for a phone-camera interface to a LED-based physical device.

The main outcomes are:

- an Android camera receiver prototype with real-time marker acquisition, LED observation extraction, preamble acquisition, decoding progress, and final message display;
- a virtual browser transmitter that can be used for protocol and UI testing before physical hardware is available;
- a modular architecture separating computer vision, timing recovery, packet sampling, and message decoding;
- a soft rateless Raptor-like transmission model based on precode protection, LDGM-style measurements, and online belief propagation;
- a receiver workflow where progress can survive short observation gaps instead of requiring a full restart;
- logging and desktop analysis tools for evaluating recorded streams and comparing receiver changes.

These results can be used in several ways. The virtual transmitter can be replaced by a physical LED device while preserving the same marker layout and packet protocol. The Android receiver can be integrated into a product prototype. The CV layer can be optimized independently for robustness and speed. The protocol decoder can be tuned using recordings from real devices and different phone models.

The most reusable outcome is the protocol architecture. It turns an unreliable five-bit visual channel into a soft, rateless measurement stream. That gives the company a path toward a practical optical receiver even when the camera-based vision layer is imperfect.

Remaining Work

The project is still in progress. The most important remaining tasks are:

- optimize the marker tracker and LED scorer for real-time mobile performance;

- test the receiver with physical hardware rather than only a virtual screen transmitter;
- tune preamble detection and clock synchronization on a larger set of recordings;
- further evaluate the online belief propagation decoder under different rates, phones, and lighting conditions;
- continue collecting datasets and logs to compare receiver changes objectively.